

Pandas Practice Questions for Finance

Beginner → Intermediate | Google Colab | Questions only — no solutions

Practice dataset

Use a DataFrame named **df** containing finance transaction data with columns such as Invoice No, Date, Supplier, Account, Quantity, UnitPrice, Debit, Credit, Budget and Actual. Use the columns available in your own dataset where appropriate.

Part A — Pandas Fundamentals

1. Import Pandas and load an Excel file into a DataFrame.
2. Display the first 5 rows.
3. Display the last 5 rows.
4. Find the number of rows and columns.
5. Display all column names.
6. Check the data type of every column.
7. Generate a statistical summary of numeric columns.
8. Select only Invoice No and Supplier.
9. Select Debit and Credit.
10. Count missing values in every column.

Part B — Missing Data & Cleaning

1. Use `fillna()` to replace missing numeric values with 0.
2. Fill missing UnitPrice values using the mean UnitPrice.
3. Use `dropna()` and compare row counts before and after.
4. Find rows where Invoice No is missing.
5. Convert a numeric-looking text column using `pd.to_numeric()`.
6. Use `errors='coerce'` and identify values converted to NaN.
7. Remove leading/trailing spaces from Supplier names.
8. Convert a text column to lowercase using `.str`.
9. Check whether Invoice No contains duplicates.
10. Display all rows belonging to duplicated invoices.

Part C — Invoice & Transaction Validation

1. Create `TransactionValue = Quantity × UnitPrice`.
2. Compare TransactionValue with the recorded Debit/Credit amount.
3. Create a Validation column showing Valid or Check.

4. Find all transactions requiring checking.
5. Count duplicated invoice numbers.
6. Find invoices appearing more than once using `value_counts()`.
7. Check whether any invoice has both Debit and Credit populated.
8. Calculate net transaction amount as Debit – Credit.
9. Find transactions with a negative net amount.
10. Create an exception report containing failed validation records.

Part D — Filtering & Sorting

1. Show transactions where Debit is greater than 50,000.
2. Show transactions where Credit is greater than 25,000.
3. Show transactions where UnitPrice is above the average.
4. Filter transactions for one supplier.
5. Show transactions where TransactionValue exceeds 100,000.
6. Sort by TransactionValue from highest to lowest.
7. Find the highest TransactionValue.
8. Find the lowest TransactionValue.
9. Display the top 5 transactions by TransactionValue.
10. Display the top 5 transactions with the largest absolute validation difference.

Part E — Grouping & Finance Analysis

1. Group by Supplier and calculate total Debit.
2. Group by Supplier and calculate total Credit.
3. Group by Account and calculate total Debit and Credit.
4. Calculate total Debit, total Credit and net balance.
5. Find the supplier with the highest total Debit.
6. Find the account with the highest total Credit.
7. Create $\text{Variance} = \text{Actual} - \text{Budget}$.
8. Calculate Variance %.
9. Identify records where Actual is above Budget.
10. Create a management summary with Budget, Actual, Variance and Variance %.

Part F — Practical Finance Challenges

1. Create an invoice exception report containing duplicates, missing values and validation failures.
2. Identify the 5 transactions requiring the most management attention and define your rule.
3. Create a supplier summary with transaction count, total Debit, total Credit and net amount.

4. Calculate the percentage of transactions that passed validation.
5. Identify accounts where total Debit differs significantly from total Credit.
6. Export the cleaned DataFrame to a new Excel file.
7. Write three management observations based only on your Pandas analysis.
8. Prepare an Accounts Payable review table with Invoice No, Supplier, TransactionValue, recorded amount and Validation.

Mini Interview Practice

1. Explain the difference between selecting one column and selecting multiple columns in Pandas.
2. What is the purpose of `fillna()`?
3. What does `pd.to_numeric()` do?
4. What is the difference between `duplicated()` and `drop_duplicates()`?
5. What does `groupby()` help you do in financial analysis?
6. What is the difference between `axis=0` and `axis=1`?
7. Why is data validation important before preparing a finance report?
8. How would you explain a Pandas duplicate-invoice check to an auditor?

Suggested workflow: Import → Inspect → Clean → Validate → Calculate → Filter → Group → Summarize → Export.